

# Schur partitions of semi-structures

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# Schur Ring

A **Schur Ring** is a partition  $S$  of a set  $G$  in which for all  $B, C \in S$

$$BC = \sum_{D \in S} nD, n \in \mathbb{Z}$$

Elements of a Schur ring are called **primitive sets**.

# Schur Ring Example

In  $\mathbb{Z}_5$  we have the following Schur rings:

**Trivial:**  $\{\{1\}, \{z, z^2, z^3, z^4\}\}$

**Discrete:**  $\{\{1\}, \{z\}, \{z^2\}, \{z^3\}, \{z^4\}\}$

**Inverses:**  $\{\{1\}, \{z, z^4\}, \{z^2, z^3\}\}$

We will verify that the inverse pairs do in fact form a Schur ring on this group.

# Schur Ring Example

When verifying that a partition is a Schur ring, it can be helpful to consider it as a group ring.

So in  $\mathbb{Z}_5$ , the Schur ring  $\{\{1\}, \{z, z^4\}, \{z^2, z^3\}\}$  behaves as follows:

$$1(z + z^4) = z + z^4 \quad 1(z^2 + z^3) = z^2 + z^3$$

$$(z + z^4)^2 = z^2 + 2(1) + z^3 = 2(1) + (z^2 + z^3)$$

$$(z^2 + z^3)^2 = z^4 + 2(1) + z = 2(1) + (z + z^4)$$

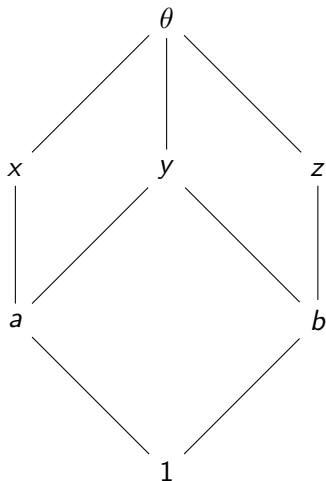
$$(z + z^4)(z^2 + z^3) = z^3 + z^4 + z + z^2 = (z^2 + z^3) + (z + z^4)$$

# Semilattice

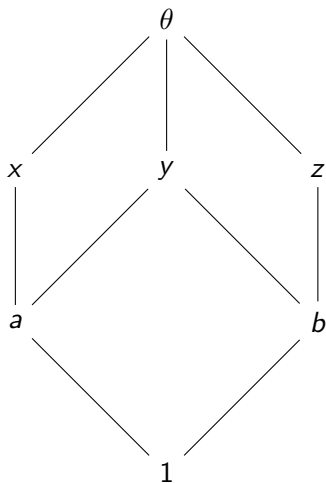
A **semilattice**  $S$  is a partially ordered set where the binary operation is that of a **least upper bound**.

# Semilattice Example

The diagram provides a semilattice.



# Semilattice Schur Ring Example



The Schur rings here are:

- 1)  $\{\{1\}, \{a, b\}, \{x, z\}, \{y\}, \{\theta\}\}$
- 2)  $\{\{1\}, \{a, b\}, \{x\}, \{y\}, \{z\}, \{\theta\}\}$
- 3)  $\{\{1\}, \{a\}, \{b\}, \{x\}, \{y\}, \{z\}, \{\theta\}\}$

# Semilattice Distinguished Element

In any finite semilattice  $S$  there exists an element  $\theta$  satisfying  $\theta \geq x$  for all  $x \in S$ . We will show that  $\theta$  cannot be in the same primitive set as any other element of the semilattice.



## $\theta$ Primitive Set

Suppose that in a Schur ring  $S$  of a semilattice  $L$  there is a primitive set  $P = \{\theta, x, \dots\}$ . Consider the square of this set, that is, consider  $(\theta + x + \dots)^2$ .

## $\theta$ Primitive Set

Consider  $(\theta + x + \dots)^2$ .

In this product we will have a minimum of  $|P|$  different  $\theta$ 's because we will have  $\theta$  times every element of  $P$  as  $\{\theta\theta, \theta x, \dots\}$ . Since  $\theta$  appeared in the product of primitive sets, we expect to get  $|P|$  copies of every other element, but we will see we cannot. The total amount of elements we need to account for is  $|P|^2$ .

## $\theta$ Primitive Set

The total amount of elements we need to account for is  $|P|^2$ . We have shown that there will be at least  $|P|$  copies of  $\theta$  and so there must be  $|P|$  copies of every other element in  $P$  generated in this product. This means we have  $|P|^2$  elements accounted for, and cannot have any more repeated elements.

## $\theta$ Primitive Set

We have already accounted for all the products  $\theta P$ , and as we consider the products  $xP$  we find that  $x\theta \in xP$ .

Since  $x\theta = \theta$  this is another  $\theta$ , but we had accounted for the maximal amount of  $\theta$  we could have. This that  $P^2$  cannot be expressed as a linear combination of primitive sets. So  $P$  cannot be a primitive set, and thus the only primitive set that can contain  $\theta$  is  $\{\theta\}$ .

# Final Theorem

## Theorem

The maximal element of any finite semilattice must be isolated in its Schur rings.